The Stark Zeeman Line-Shape Code
- User Manual and Theoretical Guide -

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Open-Source Coupled Stark-Zeeman Plasma Line-Shape Project

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1 Introduction

Modeling the spectral line shapes of multi-electron ions in hot, dense plasmas is a challenging problem in plasma physics, astrophysics, and laboratory nuclear fusion diagnostics. The presence of strong external magnetic fields ($\vec{B}$) coupled with the stochastic electric microfields ($\vec{F}$) produced by moving ions and electrons in the plasma leads to a complex, simultaneous splitting and broadening known as the Stark-Zeeman (SZ) effect.

The **Stark Zeeman Line-Shape** (**StarkZee**) code is an open-source Python implementation that couples atomic physics calculations in arbitrary magnetic fields with plasma line-broadening theories under quasi-static electric microfields and electron impact. The library incorporates ion dynamics and electric field fluctuations via the Frequency Fluctuation Model (FFM).

This manual details the theoretical background, governing equations, code structure, physical simplifications, and high-performance NumPy vectorization optimizations implemented in the package.

2 Theoretical Background and Governing Equations

2.1 Liouville Space Representation

The spectral line-shape intensity $I(\omega)$ of emitted radiation is related to the Fourier transform of the dipole autocorrelation function $C(t)$ by [2]:

$$I(\omega) = \frac{1}{\pi}\mathrm{Re}\int_0^\infty C(t)e^{i\omega t}dt \tag{1}$$

In Liouville space notation, the autocorrelation function is represented as a trace over the quantum emitter states:

$$C(t) = \langle\langle \vec{d}^* | U(t) | \vec{d}\rho_0 \rangle\rangle \tag{2}$$

where $\vec{d}$ is the electric transition dipole operator, ρ_0 is the equilibrium density matrix of the initial manifold, and $U(t)$ is the time-evolution propagator.

In the anti-symmetric subspace (ground-to-excited manifold transitions in the no-quenching approximation), the Liouvillian operator L is constructed from the excited-state Hamiltonian H_u and ground-state Hamiltonian H_l:

$$L = \frac{1}{\hbar}(H_u \otimes I^d - I \otimes H_l^d) \tag{3}$$

The diagonal elements of the Liouvillian represent the transition frequencies, and the off-diagonal elements represent the Stark couplings.

2.2 Radiator Hamiltonian in Magnetic Fields

For a radiator state with principal quantum number n and nuclear charge Z in an external magnetic field $\vec{B} = B\hat{z}$, the excited-state atomic Hamiltonian in the uncoupled basis $|n, l, m_l, s, m_s\rangle$ is given by [3]:

$$H_A = H_0 + V_{SO} + H_Z^{(1)} + H_Z^{(2)} \tag{4}$$

where:

1. **Unperturbed Hydrogenic Energy**:

$$H_0 = -\frac{Z^2\mathrm{Ry}}{n^2}I \tag{5}$$

2. **Spin-Orbit Interaction**:

$$V_{SO} = \xi \vec{L} \cdot \vec{S} \tag{6}$$

with the spin-orbit parameter:

$$\xi = \frac{Z^4 \alpha^2 \mathrm{Ry}}{n^3 l(l+1/2)(l+1)} \tag{7}$$

3. **Linear Zeeman Effect**:

$$H_Z^{(1)} = \mu_B B (L_z + g_s S_z) \tag{8}$$

where μ_B is the Bohr magneton and $g_s \approx 2.00232$.

4. **Quadratic Zeeman (Diamagnetic) Effect**:

$$H_Z^{(2)} = \frac{e^2 B^2}{8 m_e} r^2 \sin^2 \theta \tag{9}$$

2.3 Empirical Field-Free Energies

The analytic diagonal $H_0 = -Z^2\,\mathrm{Ry}_{\mathrm{red}}/n^2$, together with the spin-orbit and fine-structure terms, reproduces the field-free level positions only to the accuracy of the hydrogenic Dirac formula. For quantitative line centers, `StarkZee` can instead inject *measured* field-free energies through the `use_empirical_data=True` flag (with the element symbol `atom`). The values are read from a tabulated database (`atomic_data.load_levels`) holding NIST level energies in wavenumbers [cm^{-1}], resolved by (n, l, j). The empirical Hamiltonian is kept in cm^{-1} throughout this code path so that callers can work entirely in wavenumber units; all Zeeman contributions are also converted from eV to cm^{-1} before being added, keeping every term on the same scale.

Because the field-free Hamiltonian is degenerate (the Dirac formula makes $2s_{1/2}$ and $2p_{1/2}$ coincide), the empirical energies cannot simply be written on the diagonal of the uncoupled $|n, l, m_l, m_s\rangle$ basis: `numpy.linalg.eigh` is free to mix the degenerate l states arbitrarily. `StarkZee` therefore:

1. diagonalizes a degeneracy-broken field-free Hamiltonian (unperturbed energy + spin-orbit only, *omitting* the mass-velocity/Darwin term so that l remains a good label), yielding eigenvectors V;

2. labels each coupled eigenstate k by its dominant orbital component l and its total angular momentum $j = l \pm \frac{1}{2}$, the branch being set by the sign of $\langle \vec{L} \cdot \vec{S} \rangle = (E_k - E_n)/\xi_{nl}$;

3. assigns the tabulated energy $D_k = E^{\mathrm{emp}}(l, j)$ [cm^{-1}] to each eigenstate and reconstructs the field-free Hamiltonian as

$$H_0^{\mathrm{emp}} = V \,\mathrm{diag}(D)\, V^\dagger \quad [\mathrm{cm}^{-1}]. \tag{10}$$

The Zeeman terms ($\mu_B B(m_l + g_s m_s)$ and the diamagnetic correction) are converted from eV to cm^{-1} and added on top of H_0^{emp}. Since the tabulated values are absolute level energies (ground state at 0), transition wavenumbers follow directly as differences $E_u - E_l$ and reproduce the observed NIST Lyman/Balmer line centers.

2.4 Full Electron-Radiator Hamiltonian

Under the quasi-static ion approximation, the radiator is subjected to a constant electric microfield $\vec{F}$ at an angle θ relative to the magnetic field $\vec{B}$. The microfield is decomposed into longitudinal ($F_z = F\cos\theta$) and transverse ($F_x = F\sin\theta$) components. The Stark interaction is

$$V_E = -e\,\vec{r}\cdot\vec{F} = -e(zF_z + xF_x) = -\hat{\boldsymbol{d}}\cdot\vec{F} \tag{11}$$

where $\hat{\boldsymbol{d}} = e\vec{r}$ is the electric dipole operator. The full electron-radiator Hamiltonian is

$$H = H_A + V_E = H_0 + V_{\mathrm{SO}} + H_Z^{(1)} + H_Z^{(2)} + V_E \tag{12}$$

H_A and V_E are assembled into a single matrix and diagonalized *as a whole* by `numpy.linalg.eigh` at every microfield quadrature point (`solve_starkzee`). The Stark interaction is **not** treated perturbatively: there is no expansion in powers of V_E. This exact treatment is essential when the Stark energy $eF\langle r\rangle_n$ is comparable to the fine-structure or Zeeman splittings, yielding the Stark-dressed eigenstates $|k\rangle$ and transition frequencies ω_k.

2.5 Plasma Microfield Distributions

The electric microfield is averaged over a probability distribution $W(F)$. SZLS implements both:

1. **Holtsmark Distribution**: Standard unscreened microfield for weakly coupled plasmas [9]:

$$W_H(\beta) = \frac{2\beta}{\pi}\int_0^\infty y\sin(\beta y)e^{-y^{1.5}}dy \tag{13}$$

   where $\beta = F/F_0$ is the normalized field strength.

2. **Hooper Distribution**: Debye-screened microfield distribution [10]:

$$W_{Hooper}(\beta, a) = \frac{2\beta}{\pi}\int_0^\infty y\sin(\beta y)e^{-y^{1.5}\chi(y,a)}dy \tag{14}$$

   where $\chi(y,a) = (1 + 1.5a^2/y^2)^{-0.75}$ is the screening correction fit, and $a = r_e/\lambda_D$ is the ratio of the ion-sphere radius to the Debye length.

3. **Potekhin Distribution**: Analytical fits for electric microfield distributions $P(\beta)$ in plasmas [11], which support both screened and unscreened cases as well as neutral and charged radiator points, incorporating the ion-ion coupling parameter Γ.

2.6 Electron Impact Broadening

Fast-moving electrons are treated in the *impact* (completed-collision) approximation, contributing a homogeneous Lorentzian broadening to every transition component. The half-width at half-maximum (HWHM) γ_e is computed from the semi-classical **GBK (Griem-Baranger-Kolb)** model [8, 7]:

$$\gamma_e(\Delta E) = W_{\mathrm{pref}}\,\langle r^2\rangle_n\left[C_n + \frac{1}{2}E_1(y(\Delta E))\right] \tag{15}$$

where:

- $W_{\mathrm{pref}} = \dfrac{4\pi}{3}N_e\sqrt{\dfrac{2m_e}{\pi k_B T_e}}\left(\dfrac{\hbar}{m_e}\right)^2\dfrac{\hbar}{e}$ is the density–temperature prefactor (the $\hbar/e$ factor converts rad s^{-1} to eV).

- $\langle r^2 \rangle_n$ is the shell-averaged mean-square radius, computed exactly by quantum-mechanical summation over l-subshells:

$$\langle r^2 \rangle_n = \frac{1}{n^2} \sum_{l=0}^{n-1} (2l+1) \frac{n^2}{2Z^2} \left[5n^2 + 1 - 3l(l+1)\right] a_0^2 \tag{16}$$

  This expression scales as n^4/Z^2 and is *not* replaced by an approximate constant.

- C_n is the strong-collision term ($C_n = 1.5$ for $n \leq 2$, 0.75 for $n = 3, 4$, 0.40 for $n \geq 5$).

- $E_1(y)$ is the exponential integral, and the argument

$$y(\Delta E) = \left(\frac{n^2}{2Z}\right)^2 \frac{\Delta E^2 + E_c^2}{E_H \, T_e} \tag{17}$$

  depends on the energy detuning $\Delta E = E - E_{pk}$ [eV] from the transition center, with $E_c = \hbar\omega_c$ [eV] the cutoff energy and $E_H = 2\,\mathrm{Ry} \approx 27.211\,\mathrm{eV}$ the Hartree energy.

- The cutoff frequency $\omega_c = \max(\omega_p, \omega_L, \omega_e)$ accounts for the three physical limits of the collision trajectory: the plasma frequency ω_p, the electron Larmor frequency $\omega_L = eB/m_e$ (relevant at strong B), and the inter-electron configuration rate $\omega_e = 2\pi v_{\mathrm{th}}/r_e$.

Frequency-dependent vs. resonance-center width. By default (`frequency_dependent_width=True`), $\gamma_e(\Delta E)$ is evaluated at the actual detuning of each transition component using the full $E_1(y(\Delta E))$ function. When `frequency_dependent_width=False`, the width is fixed at the line-center value $\gamma_e(0)$, which reduces computation time at the cost of accuracy in the far wings.

Application in the profile. Each Stark-dressed transition component $(i \to j, q)$ is broadened by a Lorentzian of half-width γ_e:

$$L_{ij}(E) = \frac{\gamma_e/\pi}{(E - E_{ij})^2 + \gamma_e^2} \tag{18}$$

and the contribution is weighted by the component intensity $|d_q(i,j)|^2$ and the microfield quadrature weight.

2.7 Thermal Doppler Broadening

Thermal motion of the radiating ions causes each photon frequency to be Doppler-shifted by $\delta\omega = \omega_0 \, v_z/c$. Averaging over a Maxwell–Boltzmann velocity distribution yields a Gaussian line shape with $1/e$ half-width

$$\Delta\omega_D = \omega_0 \sqrt{\frac{2k_B T_i}{m_{\mathrm{ion}} c^2}} \quad \Longleftrightarrow \quad \Delta E_D = E_0 \sqrt{\frac{2T_i}{m_{\mathrm{ion}} c^2/e}} \tag{19}$$

where T_i is the ion temperature, m_{ion} the emitter mass, and E_0 the transition energy in eV. For H Balmer-α at $T_i = 5\,\mathrm{eV}$, this gives $\Delta E_D \approx 0.062\,\mathrm{meV}$ — comparable to the Zeeman splitting at 1 T and negligible relative to the Stark width at $N_e \sim 10^{23}\,\mathrm{m}^{-3}$.

Important: Doppler broadening is *not* applied automatically. `calculate_static_profile()` returns the purely Stark-Zeeman-broadened profile (ion quasi-static + electron impact only). Doppler broadening must be applied *after* as a post-processing step via `convolutions.py`:

- `apply_doppler_broadening(wavelengths_nm, profile, Ti_ev, species='H')` — convolves the profile with a Gaussian kernel of the thermal Doppler width.

- `apply_instrument_broadening(wavelengths_nm, profile, fwhm_nm)` — convolves with a Gaussian instrumental slit function.

Both use FFT convolution (`scipy.fft`) for efficiency. Both operate on a *uniform wavelength grid*; convert energy grids to wavelength before calling. The convolution preserves the total integrated intensity.

The complete broadening pipeline is therefore:

$$I_{\mathrm{obs}}(\lambda) = \Big[I_{\mathrm{SZ}}(\lambda) * G_D(\lambda)\Big] * G_{\mathrm{inst}}(\lambda) \tag{20}$$

where I_{SZ} is the Stark-Zeeman profile from `static_profile.py` (via `line_profile.py`), G_D is the Doppler Gaussian, and G_{inst} the instrumental Gaussian.

2.8 Frequency Fluctuation Model (FFM)

Dynamic ion motion causes the static microfield to fluctuate over time. In the FFM [6], this is modeled as a stationary Markovian process that mixes the Stark-dressed transition components at a rate:

$$\nu = \frac{v_{th}}{r_s} \tag{21}$$

where $v_{th} = \sqrt{2k_BT_i/m_i}$ is the perturber thermal speed, and r_s is the ion-sphere radius. The dynamic line profile is computed by inverting the Markovian mixing matrix:

$$I(\omega) = \frac{r^2}{\pi}\mathrm{Re}\frac{S(\omega)}{1-\nu S(\omega)} \tag{22}$$

with the static propagator sum:

$$S(\omega) = \sum_k \frac{p_k}{\nu + \gamma_e + i(\omega - \omega_k)} \tag{23}$$

where $p_k = a_k/r^2$ is the normalized transition probability of the k-th component.

3 Physical Features and Validation

3.1 Quadratic Zeeman polarization wings

At $B \geq 500\,\mathrm{T}$ the diagonal QZ shifts within $n = 3$ differ by up to $5\,\mathrm{meV}$, separating the 3p($m_l = \pm1$)→2s transitions (which carry ≈ 21% of the Hα oscillator strength and exhibit the *largest* n=3 QZ shift, +8.87 meV at $B = 1000\,\mathrm{T}$) from the dominant 3d→2p cluster by ≈ 7 meV. The resulting "wings" appear only in the `quadratic_zeeman=True` profiles and are validated in `test_12_halpha_qz_wings.py`.

3.2 $\pm2\mu_BB$ Stark-Zeeman satellite features

A transverse microfield F_x couples adjacent-m_l states within the same principal shell via $\Delta l = \pm1$, $\Delta m_l = \pm1$ matrix elements. The upper eigenstate near $E_{0,n} + 2\mu_BB$ (dominated by $|nd,\, m_l{=}2\rangle$) acquires a small $|np,\, m_l{=}1\rangle$ admixture β, producing a σ^+ transition to the zero-Zeeman lower state at photon energy $E_0 + 2\mu_BB$.

Hβ (n=4→2). The satellite is a *distinct peak* at +117 meV (≈ 2% of the main σ^+ intensity), because $n = 4$ has both $|4d,\, m_l{=}2\rangle$ and $|4f,\, m_l{=}2\rangle$ *degenerate* at $+2\mu_BB$, greatly amplifying Stark mixing. The satellite is visible in the Balmer-series profile at ≈ 465 nm and ≈ 509 nm for $B = 1000\,\mathrm{T}$.

Hα (n=3→2). Only $|3d, m_l{=}2\rangle$ sits at $+2\mu_B B$ (no $l = 3$ sub-states exist for $n = 3$). The satellite amplitude ($\approx 0.07\%$ of main peak) is comparable to the Lorentzian tail of the σ^+ main peak at that position, so no distinct local maximum appears at 618 nm / 700 nm. This result is converged with respect to the microfield grid size (`num_f` = 30, 60, 100 give identical profiles) and is validated in `test_13_stark_zeeman_satellites.py`.

B-dependence and observability. The mixing coefficient $\beta = F_x \langle np|r|nd\rangle/(\mu_B B)$ scales as B^{-1}, so β is *larger* at lower B. However, the satellite position $2\mu_B B$ also shrinks with B. At $B = 100\,\mathrm{T}$, $2\mu_B B = 11.6\,\mathrm{meV}$ falls inside the Stark-broadened σ^+ cluster ($\sim 10\,\mathrm{meV}$ width for $n = 4$, $N_e = 10^{23}\,\mathrm{m}^{-3}$). The satellite merges into the cluster tail: the peak/valley contrast in the satellite sub-window is ≈ 1.000 (indistinguishable), versus ≈ 2.26 at $B = 1000\,\mathrm{T}$ where the satellite is clearly separated. *High B is therefore required for the satellite to appear as a resolved, separated peak.*

4 Physical and Numerical Simplifications

To achieve absolute self-sufficiency, maintainability, and high execution speeds, the following simplifications were incorporated:

1. **Analytical Hydrogenic Basis**: Instead of calling external commercial databases or complex atomic structure codes (e.g., Cowan or FAC), SZLS implements exact analytical hydrogenic radial wavefunctions [3]:

$$R_{nl}(r) = \sqrt{\left(\frac{2Z}{n}\right)^3 \frac{(n-l-1)!}{2n(n+l)!}}\, e^{-Zr/n} \left(\frac{2Zr}{n}\right)^l L_{n-l-1}^{2l+1}\left(\frac{2Zr}{n}\right) \tag{24}$$

   This provides self-contained matrix elements for r and r^2 calculations.

2. **Resonance-center impact approximation (optional)**: By default (`frequency_dependent_width=Tr`
   the GBK electron width $\gamma_e(\Delta E)$ is evaluated at the actual detuning of each transition component. Setting `frequency_dependent_width=False` fixes the width at its line-center value $\gamma_e(0)$, avoiding repeated evaluations of $E_1(y)$ at every wavelength point. This is a useful approximation when the profile wings are not of primary interest.

3. **Gauss-Legendre Angle Quadrature**: Integrating over the field direction is performed using a 6-point Gauss-Legendre quadrature over the variable $\mu = \cos\theta \in [0, 1]$.

5 Built-in Reference Models

The `starkzee.models` package provides five independent lineshape models that share a common call signature and serve as benchmarks against StarkZee's fully coupled solver. They are grouped into tabulated models (reading precomputed databases) and analytical models.

5.1 Tabulated Models

5.1.1 Stehlé (MMM) — `stehle`

The Stehlé model reads precomputed Stark lineshapes from the Meudon Molecular Model (MMM) database [?] for hydrogen Balmer and Paschen transitions over a grid of N_e and T_e values. The tables contain the *unmagnetized* ($B = 0$) Stark + fine-structure profile as a function of a reduced detuning $\Delta\omega/F_0$ (Holtsmark normal field units). The pipeline is:

1. Bilinear interpolation over the (N_e, T_e) table grid (with a Fortran-style three-point hyperbolic interpolator for the detuning axis).

2. Thermal Doppler broadening via FFT convolution.
3. Normal Zeeman splitting applied *after* the table: the whole Stark+Doppler profile is rigidly copied to $\omega \pm \omega_L$ (Larmor frequency) with angle-dependent π/σ weights:

$$I(\theta) = \sin^2\theta \, I_\pi + \tfrac{1+\cos^2\theta}{2} \, (I_{\sigma+} + I_{\sigma-}) \tag{25}$$

The Stark and Zeeman effects are therefore treated as **separable**. This is valid only when the Zeeman splitting $\mu_B B$ is much smaller than the Stark width. Stehlé covers arbitrary (n_u, n_l) and species with no restriction on B-field value (it is simply ignored in the table lookup).

5.1.2 Rosato — `rosato`

The Rosato database [?] solves the Stark-Zeeman problem *jointly* and tabulates the result with B as an explicit table axis ($B \in \{0, 1, 2, 2.5, 3, 5\}$ T). The pipeline is:

1. Interpolation over the (N_e, T_e, B) grid. The B-interpolation is *scaled*: before blending two bracketing profiles at B_0 and B_1, each profile's detuning axis is stretched by B_{node}/B. This exploits the fact that Zeeman splitting scales linearly with B, aligning the σ components before interpolating.
2. Angle dependence from *two real tables* (parallel and perpendicular to $\vec{B}$), blended as $I = I_{\|} \cos^2\theta + I_{\perp} \sin^2\theta$. Both tables carry the correct π/σ lineshape and width — the angle enters as a physical interpolation, not just an amplitude weight.
3. Thermal Doppler broadening via FFT convolution.

Rosato is restricted to deuterium Balmer lines and the table coverage $N_e \in [10^{13}, 10^{16}]$ cm^{-3}, $T_e \in [0.316, 31.6]$ eV, $B \leq 5$ T.

Stehlé vs. Rosato. The central physical distinction is in the treatment of $\vec{B}$:

	Stehlé	**Rosato**
Table contents	Field-free Stark only	Stark + Zeeman jointly
B enters as	Post-processing rigid triplet	Real table axis
Angle dependence	Analytic π/σ weights	Two precomputed angle tables
Stark–Zeeman coupling	None (separable approx.)	Fully captured
Coverage	Arbitrary (n_u, n_l), all B	D Balmer only, $B \leq 5$ T

Rosato is physically correct precisely where Stehlé's separability assumption breaks down: when the Zeeman splitting $\mu_B B$ is comparable to or larger than the Stark width. For Hα at $B = 10$ T, $\mu_B B \approx 0.58$ meV, while the Stark width at $N_e = 5 \times 10^{21}$ m^{-3} is ≈ 0.8 meV — the same order of magnitude, making the Stark–Zeeman coupling non-negligible.

5.2 Analytical Models

5.2.1 Lomanowski — `lomanowski`

Polynomial fits for the Stark FWHM from Lomanowski *et al.* [?]:

$$\Delta\lambda_S = c_{n_u n_l} \, N_e^{a_{n_u n_l}} \, T_e^{-b_{n_u n_l}} \quad [\text{nm}] \tag{26}$$

Coefficients (a, b, c) are tabulated for H/D Balmer and Paschen lines up to $n_u = 9$. A Voigt profile combines this Stark Lorentzian with a Doppler Gaussian. No magnetic field treatment.

5.2.2 Parameterized Stehlé — `stehle_param`

An analytical parametric fit to the Stehlé tables, providing a fast closed-form approximation to the field-free Stark profile without reading the NetCDF database.

5.2.3 Voigt — `voigt`

A pseudo-Voigt profile using the Griem α_{12} Stark half-width [7] combined with thermal Doppler broadening. Serves as a quick estimate when only order-of-magnitude Stark width accuracy is required.

5.3 Comparison with StarkZee

Feature	**StarkZee**	**Stehlé**
Magnetic field	Full simultaneous diagonalization of $H = H_A + V_E$	$B = 0$ in tables; added as triplet
Fine structure	Yes — spin-orbit, MV, Darwin	No (degenerate hydrogenic)
Electron broadening	GBK semi-classical, frequency-dependent	Unified theory (more rigorous fa
Ion dynamics	Quasi-static + optional FFM	Static + some ion-dynamics trea
Microfield	Hooper screened	Holtsmark/Hooper variant
B-treatment	Intrinsic to Hamiltonian	External convolution

The most important physical distinction at $B \neq 0$: StarkZee diagonalizes $H = H_A + V_E$ *simultaneously* for all field strengths. When $\mu_B B \sim 3n\, e\, a_0\, F$ (Zeeman and Stark splitting comparable) the eigenstates are genuine Stark-Zeeman hybrids — neither pure Zeeman nor pure Stark states. No post-processing convolution can reproduce this mixing. The separable approximation (Stehlé) and scaled-interpolation approach (Rosato) both become inaccurate in this regime.

6 Code Structure and Modular Architecture

The **StarkZee** package (`starkzee/`) is organized as a modular Python library:

- `starkzee/line_profile.py`: High-level `LineProfile` class. Manages spectral grids, unit conversions, Doppler broadening, and exposes all results as attributes. Entry point for most users.

- `starkzee/static_profile.py`: Core solver. `calculate_static_profile()` runs the microfield quadrature loop and accumulates the three polarization profiles P_π, P_{σ^+}, P_{σ^-}.

- `starkzee/atomic_hamiltonian.py`: Constructs H_A — the atomic Hamiltonian including unperturbed energy, spin-orbit, fine-structure, and linear and quadratic Zeeman terms — and builds the dipole operator matrices using analytical angular matrix elements. H_A is *not* diagonalized here; the full $H = H_A + V_E$ is diagonalized in `solve_starkzee`.

- `starkzee/ffm.py`: Implements dynamic ion-field fluctuations via the Frequency Fluctuation Model, supporting both an analytical Sherman-Morrison solver and a full matrix inversion.

- `starkzee/microfield.py`: Generates screened Hooper and unscreened Holtsmark microfield distributions; supports multi-species Debye lengths and custom tabulated distributions.

- `starkzee/broadening.py`: GBK electron-impact half-width $\gamma_e(\Delta E)$; computes plasma, Larmor, and configuration-change cutoff frequencies.

- `starkzee/convolutions.py`: FFT-based post-processing convolutions. `apply_doppler_broadening(wav profile, Ti_ev, species)` and `apply_instrument_broadening(wavelengths_nm, profile, fwhm_nm)` both operate on uniform wavelength grids:

$$I_{\text{conv}}(\lambda) = \mathcal{F}^{-1}\{\mathcal{F}[I(\lambda)] \cdot \mathcal{F}[K_{\text{Doppler}}(\lambda)] \cdot \mathcal{F}[K_{\text{slit}}(\lambda)]\} \tag{27}$$

- `starkzee/models/`: Comparison lineshape models (voigt, lomanowski, stehle_param, stehle, rosato) sharing a common signature, centered on NIST air wavelengths for each transition [12].

- `starkzee/atomic_loader.py`: Placeholder for future multi-electron species support.

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A Algorithmic Overview of the Calculation Pipeline

When a user invokes `LineProfile.compute_profile(grid, grid_type)`, the `StarkZee` package performs a series of nested calculations to solve the coupled Stark-Zeeman line-shape problem. The numerical delegation chain is structured as follows:

1. `LineProfile.compute_profile()`: Manages the spectral coordinate axes, performs grid unit conversions, and forwards the arguments to the profile calculations.

2. `static_profile.calculate_static_profile()`: Generates the plasma microfield grid and weights, computes the Gauss-Legendre quadrature points for the microfield orientation angle θ, and runs the main double-integration loop.

3. `solve_starkzee()`: Assembles the full electron-radiator Hamiltonian $H = H_A + V_E$ (Eq. 12) and diagonalizes it as a single matrix with `numpy.linalg.eigh`. V_E is *not* applied perturbatively.

4. `atomic_hamiltonian.build_hamiltonian()`: Builds H_A — the field-free plus magnetic Hamiltonian (unperturbed energy, spin-orbit, fine structure, linear and quadratic Zeeman) in the uncoupled $|n, l, m_l, m_s\rangle$ basis.

5. `static_profile._stark_templates()`: Returns the field-independent Stark coupling matrices M_z and M_x [eV/(V/m)], cached per (n, Z), so that $V_E = F_z M_z + F_x M_x$ is formed at each quadrature point without rebuilding the matrix loop.

6. **Transition Dipole Rotation**: Rotates the transition dipole operator matrices D_q to the mixed eigenstate bases.

7. **Electron-Impact Broadening**: Evaluates the frequency-dependent electron Stark half-width $\gamma_e(\Delta E)$ via the Griem-Baranger-Kolb (GBK) model and accumulates Lorentzian profiles.

8. `convolutions.apply_doppler_broadening()`: Performs a post-processing Fast Fourier Transform (FFT) convolution to apply thermal Doppler broadening.

B Step-by-Step Mathematical Formulations

B.1 Plasma Microfield Quadrature Sampling

In a plasma, the radiating atom/ion is surrounded by a distribution of slow-moving ions that generate a fluctuating electric microfield $\vec{F}$. In the quasi-static ion approximation, the microfield is assumed to be constant during the radiative transition lifetime. The overall line profile is integrated over the probability distribution $W(\vec{F})\, d\vec{F}$.

The presence of the external magnetic field $\vec{B}$ (defining the z-axis) breaks the spatial spherical symmetry of the plasma. Consequently, the microfield integration must be performed in two dimensions: over the field magnitude $F = |\vec{F}|$ and the cosine of the angle $\mu = \cos\theta$ between $\vec{F}$ and $\vec{B}$.

B.1.1 The Hooper Screened Microfield Distribution

To account for the screening of ion Coulomb fields by the surrounding plasma electrons, `StarkZee` implements the **Hooper microfield distribution** $W(\beta, a)$, formulated via the Fourier-Laplace transform of the screened collective field:

$$W(\beta, a) = \frac{2\beta}{\pi} \int_0^\infty y \sin(\beta y)\, T(y, a)\, dy \tag{28}$$

where:

- $\beta = F/F_0$ is the reduced electric field strength.
- F_0 is the **Holtsmark normal field strength**, defined as the Coulomb field of a single perturber at the mean inter-particle distance r_e:

$$F_0 = \frac{e}{4\pi\varepsilon_0 r_e^2}, \qquad r_e = \left(\frac{3}{4\pi N_e}\right)^{1/3} \tag{29}$$

- $a = r_e/\lambda_D$ is the Debye screening parameter, where λ_D is the multi-species Debye length:

$$\lambda_D = \sqrt{\frac{\varepsilon_0 k_B T_e}{N_e e^2 \left(1 + \sum_i X_i Z_i^2\right)}} \tag{30}$$

with $X_i = N_i/N_e$ representing the fractional ion concentration.

- $T(y, a)$ is the screened characteristic function, approximated analytically by Hooper as:

$$T(y, a) = \exp\Big(-y^{3/2} S(y, a)\Big) \tag{31}$$

with the Debye screening factor:

$$S(y, a) = \left(1 + \frac{1.5\, a^2}{y^2}\right)^{-3/4} \tag{32}$$

Note: If screening is disabled (`use_screening=False`), the parameter $a = 0$ is passed, yielding $S(y, 0) = 1$ and $T(y, 0) = \exp(-y^{3/2})$, which recovers the classical unscreened **Holtsmark distribution** $W_H(\beta)$.

B.1.2 The Potekhin Microfield Distribution

To provide Zest-compatible microfield distributions, `StarkZee` includes native implementations of the analytical fits proposed by Potekhin et al. (2002) [11]. These fits describe the cumulative distribution function $Q(\beta)$ and probability density $P(\beta)$ across both screened ($s > 0$) and unscreened ($s = 0$) potentials, for neutral and charged radiators, taking into account the ion-ion coupling parameter Γ.

For the unscreened neutral point ($s = 0$), the cumulative distribution is:

$$Q(\beta) = \frac{q_0\beta^3 - 1.33\beta^{9/2} + \beta^6}{q_1 + q_2\beta^2 + q_3\beta^3 - \frac{1}{3}\beta^{9/2} + \beta^6} \tag{33}$$

where q_n are fitting parameters dependent on Γ.

For the screened charged point ($s > 0$), the probability density is fit by:

$$P(\beta) \approx \beta^2 S_N \left[A e^{-a\beta^\alpha} + B e^{-b\beta^\gamma} + \frac{e^{-\Gamma\beta^{1/2}}}{1 + c\beta^{9/2}}\right] \tag{34}$$

where S_N is the normalization constant, and parameters $A, a, \alpha, B, b, \gamma, c$ are fitting functions of s and Γ.

B.1.3 Quadrature Discretization

The double integral $\iint W(F, \mu)\, dF\, d\mu$ is discretized as follows:

1. **Field Magnitude** F: A uniform linear grid of N_F points (default 20) is established in reduced field space $\beta_i \in [0, \beta_{\max}]$ (where $\beta_{\max} = 10$). The field magnitudes and weights are:
$$F_i = \beta_i F_0, \qquad W_{F,i} = W(\beta_i, a)\, \Delta\beta \tag{35}$$
where the weights are normalized such that $\sum_i W_{F,i} = 1$.

2. **Field Cosine** μ: Integrated over $[0, 1]$ using an N_μ-point **Gauss-Legendre quadrature** (default 6). The Legendre roots $x_j \in [-1, 1]$ and weights w_j are mapped to the physical interval $[0, 1]$:
$$\mu_j = \frac{x_j + 1}{2}, \qquad W_{\mu,j} = \frac{w_j}{2} \tag{36}$$

3. **Combined Quadrature Weight**: The joint probability weight for the microfield configuration (F_i, μ_j) is:
$$W_{ij} = W_{F,i}\, W_{\mu,j} \tag{37}$$

For each active quadrature point where $W_{ij} > 10^{-15}$, the electric field is decomposed into longitudinal and transverse components relative to $\vec{B}$:

$$F_{z,ij} = F_i\, \mu_j, \qquad F_{x,ij} = F_i \sqrt{1 - \mu_j^2} \tag{38}$$

B.2 Construction of the Atomic/Magnetic Hamiltonian

For each principal quantum number shell n, `StarkZee` constructs the atomic Hamiltonian H_{atom} in the canonical uncoupled hydrogenic basis $|n, l, m_l, s = \frac{1}{2}, m_s\rangle$ of dimension $2n^2$. The uncoupled basis states are ordered by an outer loop over $l \in [0, n-1]$, an intermediate loop over $m_l \in [-l, l]$, and an inner loop over $m_s \in \{+\frac{1}{2}, -\frac{1}{2}\}$.

The atomic Hamiltonian includes five separate physical contributions:

$$H_{\text{atom}} = H_0 + H_{\text{SO}} + H_{\text{FS}} + H_Z^{(1)} + H_Z^{(2)} \tag{39}$$

B.2.1 Unperturbed Hydrogenic Energy (H_0)

The diagonal unperturbed electronic energy of shell n is corrected for the finite nuclear mass to align precisely with observed physical wavelengths:

$$\langle n, l_i, m_{l,i}, m_{s,i} | H_0 | n, l_i, m_{l,i}, m_{s,i} \rangle = -\frac{Z^2\, \text{Ry}_{\text{red}}}{n^2} \tag{40}$$

where the reduced-mass Rydberg constant is:

$$\text{Ry}_{\text{red}} = \text{Ry}_\infty \frac{M_{\text{nuc}}}{m_e + M_{\text{nuc}}} \tag{41}$$

with A_{emitter} representing the atomic mass number of the emitting isotope.

When `use_empirical_data=True`, this analytic diagonal is replaced by NIST tabulated field-free energies [cm^{-1}] projected onto the coupled eigenstate basis, and all Zeeman terms are also expressed in cm^{-1} for consistency (see Section 2.3).

B.2.2 Spin-Orbit Coupling (H_{SO})

The spin-orbit interaction $H_{\mathrm{SO}} = \xi_{nl}\, \vec{L} \cdot \vec{S}$ couples the orbital and spin angular momenta. The coupling parameter for $l > 0$ is:

$$\xi_{nl} = \frac{Z^4 \alpha^2\, \mathrm{Ry}_\infty}{n^3\, l\, (l + \frac{1}{2})\, (l + 1)} \tag{42}$$

Using angular momentum ladder operators:

$$\vec{L} \cdot \vec{S} = L_z S_z + \tfrac{1}{2}\big(L_+ S_- + L_- S_+\big) \tag{43}$$

yielding matrix elements within a given l-subshell:

- **Diagonal** ($\Delta m_l = 0$, $\Delta m_s = 0$):

$$\langle l, m_l, m_s | H_{\mathrm{SO}} | l, m_l, m_s \rangle = \xi_{nl}\, m_l\, m_s \tag{44}$$

- **Off-diagonal** ($\Delta m_l = \pm 1$, $\Delta m_s = \mp 1$): since $s = \frac{1}{2}$ the spin ladder factor equals 1 for all allowed spin-flip transitions, giving:

$$\langle l, m_l \pm 1, m_s \mp 1 | H_{\mathrm{SO}} | l, m_l, m_s \rangle = \tfrac{1}{2}\, \xi_{nl}\, \sqrt{l(l+1) - m_l(m_l \pm 1)} \tag{45}$$

B.2.3 Fine-Structure Relativistic Corrections (H_{FS})

To restore the Dirac degeneracy (e.g. $2s_{1/2}$ and $2p_{1/2}$), `StarkZee` adds the mass-velocity and Darwin terms [3], which are diagonal in $|n, l, m_l, m_s\rangle$ and independent of m_l and m_s:

$$\langle l | H_{\mathrm{FS}} | l \rangle = \begin{cases} -A_{\mathrm{FS}}\big(n - \frac{3}{4}\big) & l = 0 \\ -A_{\mathrm{FS}}\left(\dfrac{n}{l + \frac{1}{2}} - \frac{3}{4}\right) & l > 0 \end{cases} \tag{46}$$

where $A_{\mathrm{FS}} = Z^4 \alpha^2\, \mathrm{Ry}_\infty / n^4$. It is an algebraic identity that $H_{\mathrm{SO}} + H_{\mathrm{FS}}$ reproduces the exact Dirac fine-structure eigenvalues for both $j = l \pm \frac{1}{2}$.

B.2.4 Linear Zeeman Effect ($H_Z^{(1)}$)

$$\langle m_l, m_s | H_Z^{(1)} | m_l, m_s \rangle = \mu_B B (m_l + g_s m_s) \tag{47}$$

where $\mu_B \approx 5.788 \times 10^{-5}\, \mathrm{eV/T}$ and $g_s \approx 2.0023192$.

B.2.5 Quadratic (Diamagnetic) Zeeman Effect ($H_Z^{(2)}$)

$$H_Z^{(2)} = \frac{e^2 B^2}{8 m_e}\, r^2 \sin^2\theta \tag{48}$$

This operator connects states with $\Delta l = 0$ and $\Delta l = \pm 2$. Its matrix elements factor into radial and angular parts:

$$\langle l_1, m_l | H_Z^{(2)} | l_2, m_l \rangle = \left(\frac{e^2 B^2 a_0^2}{8 m_e}\right) \langle n, l_1 | r^2 | n, l_2 \rangle\, \langle l_1, m_l | \sin^2\theta | l_2, m_l \rangle \tag{49}$$

1. **Radial integrals**:

   - Diagonal ($l_1 = l_2 = l$): $\langle n, l | r^2 | n, l \rangle = \dfrac{n^2}{2Z^2}\big[5n^2 + 1 - 3l(l+1)\big]\ \ [a_0^2]$

- Off-diagonal ($l_2 = l_1 \pm 2$): computed numerically via $\int_0^\infty R_{nl_1} r^4 R_{nl_2}\, dr$, cached with `lru_cache`.

2. **Angular integrals**:
   - Diagonal: for $l = 0$, $\langle \sin^2\theta \rangle = 2/3$; for $l > 0$,

$$\langle l, m_l | \sin^2\theta | l, m_l \rangle = 1 - \frac{2l^2 + 2l - 1 - 2m_l^2}{(2l-1)(2l+3)} \tag{50}$$

   - Off-diagonal ($l_2 = l_1 + 2$, $l_{\text{low}} = \min(l_1, l_2)$): the 1 in $1 - \cos^2\theta$ vanishes for orthogonal harmonics, leaving:

$$\langle l_1, m_l | \sin^2\theta | l_2, m_l \rangle = -\sqrt{\frac{\left[(l_{\text{low}}+1)^2 - m_l^2\right]\left[(l_{\text{low}}+2)^2 - m_l^2\right]}{(2l_{\text{low}}+1)(2l_{\text{low}}+3)^2(2l_{\text{low}}+5)}} \tag{51}$$

B.3 Construction of the Static Stark Perturbation Matrix

Under the quasi-static microfield the Stark potential is:

$$V_E = -e\,\vec{r}\cdot\vec{F} = -e(zF_z + xF_x) \tag{52}$$

The selection rule is $\Delta n = 0$, $|l_i - l_j| = 1$, $\Delta m_s = 0$. In energy units [eV]:

$$\langle n, l_i, m_{l,i} | V_E | n, l_j, m_{l,j} \rangle = -\,\delta_{m_{s,i}, m_{s,j}}\ \langle n, l_i | r | n, l_j \rangle\, \langle l_i, m_{l,i} | (zF_z + xF_x)/r | l_j, m_{l,j} \rangle \tag{53}$$

B.3.1 Radial Matrix Element

$$\langle n, l | r | n, l-1 \rangle = \frac{3n}{2Z}\sqrt{n^2 - l^2} \quad [a_0] \tag{54}$$

where $l = \max(l_i, l_j)$.

B.3.2 Angular Matrix Elements

$$\frac{z}{r} = T_0^{(1)}, \qquad \frac{x}{r} = \frac{T_{-1}^{(1)} + T_{+1}^{(1)}}{\sqrt{2}} \tag{55}$$

- $q = 0$ ($\Delta m_l = 0$): $\langle l, m_l | \cos\theta | l+1, m_l \rangle = \sqrt{\dfrac{(l+1)^2 - m_l^2}{(2l+1)(2l+3)}}$
- $q = +1$ ($\Delta m_l = -1$, i.e. $m_{l,\text{low}} = m_{l,\text{up}} + 1$):

$$\langle l_1, m_{l,1} | T_{+1}^{(1)} | l_2, m_{l,2} \rangle = \begin{cases} +\sqrt{\dfrac{(l - m_{l,2})(l - m_{l,2} - 1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ -\sqrt{\dfrac{(l + m_{l,2})(l + m_{l,2} + 1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \tag{56}$$

- $q = -1$ ($\Delta m_l = +1$, i.e. $m_{l,\text{up}} = m_{l,\text{low}} + 1$):

$$\langle l_1, m_{l,1} | T_{-1}^{(1)} | l_2, m_{l,2} \rangle = \begin{cases} -\sqrt{\dfrac{(l + m_{l,2})(l + m_{l,2} - 1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ +\sqrt{\dfrac{(l - m_{l,2})(l - m_{l,2} + 1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \tag{57}$$

B.4 Construction of the Uncoupled Transition Dipole Operator

The transition dipole operator matrices D_q (for $q \in \{0, \pm 1\}$) are built in the uncoupled basis $|n, l, m_l, s, m_s\rangle$. The selection rules for the dipole transitions require $\Delta l = \pm 1$, $\Delta m_l = q$, and $\Delta m_s = 0$.

The matrix elements of D_q are factored into radial and angular parts:

$$\langle n_l, l_l, m_{l,l}, m_{s,l}|D_q|n_u, l_u, m_{l,u}, m_{s,u}\rangle = \delta_{m_{s,l}, m_{s,u}} \langle n_l, l_l|r|n_u, l_u\rangle \langle l_l, m_{l,l}|T_q^{(1)}|l_u, m_{l,u}\rangle \tag{58}$$

B.4.1 Angular Dipole Matrix Elements

The angular parts $\langle l_l, m_{l,l}|T_q^{(1)}|l_u, m_{l,u}\rangle$ correspond to the spherical components of $\vec{r}/r$ and follow from the Wigner-Eckart theorem [5]:

- For $q = 0$ ($\Delta m_l = 0$):

$$\langle l, m_l|T_0^{(1)}|l+1, m_l\rangle = \sqrt{\frac{(l+1)^2 - m_l^2}{(2l+1)(2l+3)}} \tag{59}$$

- For $q = +1$ ($\Delta m_l = -1$, branch-dependent):

$$\langle l_1, m_{l,1}|T_{+1}^{(1)}|l_2, m_{l,2}\rangle = \begin{cases} +\sqrt{\dfrac{(l-m_{l,2})(l-m_{l,2}-1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ -\sqrt{\dfrac{(l+m_{l,2})(l+m_{l,2}+1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \tag{60}$$

  where $l = \max(l_1, l_2)$.

- For $q = -1$ ($\Delta m_l = +1$, branch-dependent):

$$\langle l_1, m_{l,1}|T_{-1}^{(1)}|l_2, m_{l,2}\rangle = \begin{cases} -\sqrt{\dfrac{(l+m_{l,2})(l+m_{l,2}-1)}{2(2l-1)(2l+1)}} & l_1 = l_2 - 1 \\ +\sqrt{\dfrac{(l-m_{l,2})(l-m_{l,2}+1)}{2(2l-1)(2l+1)}} & l_1 = l_2 + 1 \end{cases} \tag{61}$$

  where $l = \max(l_1, l_2)$.

B.4.2 Hydrogenic Radial Dipole Integrals (Gordon's Formula)

For $n_u \neq n_l$, the radial transition matrix elements $\langle n_l, l_l|r|n_u, l_u\rangle$ are evaluated exactly using Gordon's (1929) analytical formula [4]:

$$\langle n_l, l_l|r|n_u, l_u\rangle = \frac{1}{Z} \frac{(-1)^{n_l - L}}{4(2L-1)!} \sqrt{\frac{(n_u+L)!(n_l+L-1)!}{(n_u-L-1)!(n_l-L)!}} \frac{(4n_u n_l)^{L+1}(n_u-n_l)^{n_u+n_l-2L-2}}{(n_u+n_l)^{n_u+n_l}} \left[F_1 - \left(\frac{n_u-n_l}{n_u+n_l}\right)^2 F_2\right] \tag{62}$$

where $L = \max(l_u, l_l)$, and F_1, F_2 are terminating Gauss hypergeometric functions ${}_2F_1$ defined as:

$$F_1 = {}_2F_1\left(-n_u + L + 1, -n_l + L, 2L, -\frac{4n_u n_l}{(n_u-n_l)^2}\right) \tag{63}$$

$$F_2 = {}_2F_1\left(-n_u + L - 1, -n_l + L, 2L, -\frac{4n_u n_l}{(n_u-n_l)^2}\right) \tag{64}$$

Because the first argument in both hypergeometric functions is a non-positive integer, the series terminates exactly as a finite sum:

$$ {}_2F_1(-m, b, c, x) = \sum_{k=0}^{m} \frac{(-m)_k (b)_k}{(c)_k k!} x^k \tag{65}$$

where $(-m)_k$ is the Pochhammer symbol for the falling factorial. This exact evaluation is free of integration cutoff errors and is cached for maximum efficiency.

B.5 Diagonalization and Transition Dipole Basis Rotation

At each quadrature point (F_i, μ_j) the total Hamiltonian is constructed for both shells:

$$H_u = H_{\mathrm{atom},u} + V_{E,u}(F_{z,ij}, F_{x,ij}), \qquad H_l = H_{\mathrm{atom},l} + V_{E,l}(F_{z,ij}, F_{x,ij}) \tag{66}$$

Diagonalization via `numpy.linalg.eigh` gives:

$$V_u^\dagger H_u V_u = \mathrm{diag}(E_u^p), \qquad V_l^\dagger H_l V_l = \mathrm{diag}(E_l^k) \tag{67}$$

The transition energy for each pair $(p \to k)$ is $E_{pk} = E_u^p - E_l^k$.

The uncoupled dipole matrices D_q (shape $2n_l^2 \times 2n_u^2$) are rotated into the mixed eigenstate bases:

$$D'_q = V_l^\dagger\, D_q\, V_u, \qquad S_q(p,k) = |D'_q(k,p)|^2 \tag{68}$$

B.6 Homogeneous Electron-Impact Broadening (GBK Model)

The electron-impact HWHM $\gamma_e(\Delta E)$ is evaluated via the semi-classical GBK model:

$$\gamma_e(\Delta E) = W_0\, \langle r^2 \rangle_{n_u} [C_{n_u} + G(\Delta E)] \tag{69}$$

B.6.1 Density-Temperature Prefactor

$$W_0 = \frac{4\pi}{3}\, N_e \sqrt{\frac{2m_e}{\pi k_B T_e}} \left(\frac{\hbar}{m_e}\right)^2 \frac{\hbar}{e} \quad [\mathrm{eV}\, a_0^{-2}] \tag{70}$$

B.6.2 Shell-Averaged Mean-Square Radius

$$\langle r^2 \rangle_{n_u} = \frac{1}{n_u^2} \sum_{l=0}^{n_u - 1} (2l+1)\, \frac{n_u^2}{2Z^2} \left[5n_u^2 + 1 - 3l(l+1)\right] \quad [a_0^2] \tag{71}$$

B.6.3 Strong-Collision Constant

$$C_{n_u} = \begin{cases} 1.50 & n_u \le 2 \\ 0.75 & n_u \in \{3, 4\} \\ 0.40 & n_u \ge 5 \end{cases} \tag{72}$$

B.6.4 Dynamical GBK Factor

$$G(\Delta E) = \frac{1}{2}\, E_1(y(\Delta E))\,, \qquad y(\Delta E) = \left(\frac{n_u^2}{2Z}\right)^2 \frac{\Delta E^2 + E_c^2}{E_H\, T_e} \tag{73}$$

where all quantities are in eV: $\Delta E = E - E_{pk}$ is the energy detuning, $E_c = \hbar\omega_c$ is the cutoff energy, $E_H = 2\,\mathrm{Ry}_\infty \approx 27.211\,\mathrm{eV}$ is the Hartree energy, and T_e is the electron temperature in eV. The cutoff $\omega_c = \max(\omega_p, \omega_L, \omega_e)$ with:

$$\omega_p = \sqrt{\frac{N_e e^2}{\varepsilon_0 m_e}}, \qquad \omega_L = \frac{eB}{m_e}, \qquad \omega_e = \frac{2\pi v_{\mathrm{th},e}}{r_e} = 2\pi \sqrt{\frac{k_B T_e}{m_e}} \left(\frac{4\pi N_e}{3}\right)^{1/3} \tag{74}$$

B.7 Profile Accumulation

Each transition is broadened by a Lorentzian of HWHM $\gamma_e(\Delta E)$:

$$L_{pk}(\Delta E) = \frac{1}{\pi} \frac{\gamma_e(\Delta E)}{\Delta E^2 + \gamma_e(\Delta E)^2} \tag{75}$$

The three polarization profiles are accumulated by vectorized summation over all quadrature weights and all Stark-dressed transitions:

$$P_q(E) = \sum_i W_{F,i} \sum_j W_{\mu,j} \sum_{p,k} S_q(p,k)\, L_{pk}(E - E_{pk}) \tag{76}$$

where $q = 0 \to P_\pi$, $q = -1 \to P_{\sigma^+}$ (blue component), $q = +1 \to P_{\sigma^-}$ (red component).

B.8 Post-Processing Thermal Doppler Broadening

Thermal motion of the radiating ions induces a Gaussian broadening with $1/e$ half-width:

$$\Delta E_D = E_0 \sqrt{\frac{2T_i}{m_{\text{ion}} c^2/e}}, \qquad \Delta\lambda_D = \lambda_0 \sqrt{\frac{2T_i}{m_{\text{ion}} c^2/e}} \tag{77}$$

where $m_{\text{ion}} = A_{\text{emitter}}\, m_p$ is the emitter rest mass, E_0 the transition energy in eV, and λ_0 the grid midpoint. The Gaussian kernel is:

$$K_D(\lambda) = \exp\left(-\frac{(\lambda - \lambda_0)^2}{\Delta\lambda_D^2} \right) \tag{78}$$

Convolution is performed via the FFT theorem on a wavelength grid edge-padded by $N/2$ on each side:

$$P_{q,\text{conv}}(\lambda) = \mathcal{F}^{-1}[\mathcal{F}(P_{q,\text{pad}}) \cdot \mathcal{F}(\text{fftshift}(K_{D,\text{pad}}))] \tag{79}$$

C Physical Approximations and Domains of Validity

Approximation	Mathematical Representation	Physical Domain of Validity	Failure Mode
Within-Shell Isolation ($\Delta n = 0$)	$V_E = \text{diag}_n(V_E)$ (no $n \to n \pm 1$ coupling)	Stark shift $\ll$ shell spacing $2Z^2\text{Ry}/n^3$	Quadratic Stark effect; Inglis-Teller line merging at $N_e \gtrsim 10^{24}\,\text{m}^{-3}$.
Quasi-Static Ion Microfields	$\vec{F}_{\text{ion}} = \text{constant}$	Perturber fluctuation rate $\nu_i \ll$ Stark width ΔE_S	Ion dynamics / motional narrowing; corrected by the FFM module.
Semi-Classical Electron Impact	Lorentzian shape with GBK width γ_e	$\lambda_{dB} \ll$ impact parameter ρ	Quantum close-coupling needed for cold dense plasmas.
Dipole Approximation	$V_{\text{rad}} \propto \vec{r} \cdot \vec{E}$	$\lambda \gg \langle r \rangle_n \sim n^2 a_0/Z$	Quadrupole / octupole transitions for high-n Rydberg states.

Table 1: Key physical approximations, their ranges of validity, and failure modes.

D Physics Explanation of Use Cases and Operational Limits

D.1 The High-Magnetic-Field Regime

In laboratory magnetic confinement fusion devices (tokamaks, stellarators), the magnetic field reaches $B \approx 1$–$10\,\text{T}$:

- The Zeeman splitting $\Delta E_Z \approx 0.1\,\mathrm{meV}$ is comparable to the fine-structure splitting of lower shells, requiring the full coupled Hamiltonian diagonalization.
- Typical microfield strengths $F \sim 10^5$–$10^6\,\mathrm{V/m}$ represent a weak-Stark regime where Stark broadening acts as a symmetric perturbation on the Zeeman triplet structure.

For astrophysical compact objects (magnetized white dwarfs with $B \sim 10^3$–$10^5\,\mathrm{T}$, neutron stars with $B \sim 10^8\,\mathrm{T}$):

- The quadratic Zeeman term $H_Z^{(2)} \propto B^2$ dominates, causing significant blue-shifting and highly asymmetric splitting. The `quadratic_zeeman=True` flag enables exact numerical treatment of the $\Delta l = \pm 2$ coupling (see Appendix B).
- `StarkZee`'s exact numerical computation of $\langle n, l_1 | r^2 | n, l_2 \rangle$ avoids the geometric-mean overestimation of up to 41% for $n = 5$.

D.2 Density Limits

- **Low density** ($N_e \lesssim 10^{18}\,\mathrm{m}^{-3}$): The profile is dominated by thermal Doppler broadening; microfield weights converge to a delta function at $F = 0$.
- **High density** ($N_e \gtrsim 10^{24}\,\mathrm{m}^{-3}$): The inter-particle spacing r_e approaches the atomic radius $\langle r \rangle_n$. The static and $\Delta n = 0$ approximations break down as the Stark shift exceeds the Rydberg shell spacing (Inglis-Teller limit).